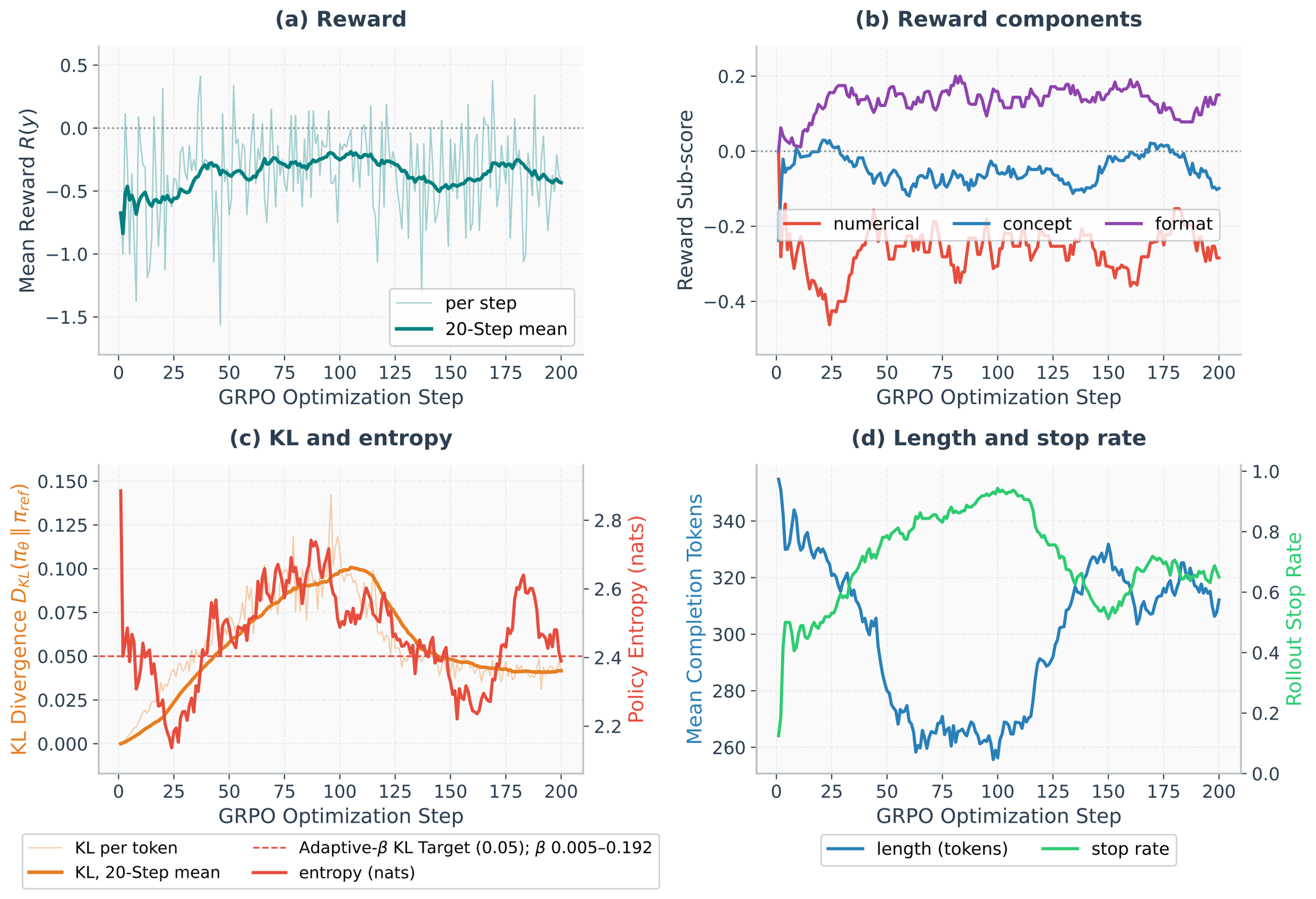
(a) Reward
Mean Reward $R(y)$
0.5
0.0
−0.5
−1.0
−1.5
per step
20-Step mean
0
25
50
75
100
125
150
175
200
GRPO Optimization Step
(b) Reward components
Reward Sub-score
0.2
0.0
−0.2
−0.4
numerical
concept
format
GRPO Optimization Step
(c) KL and entropy
KL Divergence $D_{KL}(\pi_\theta \| \pi_{ref})$
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Policy Entropy (nats)
2.8
2.6
2.4
2.2
GRPO Optimization Step
KL per token
KL, 20-Step mean
Adaptive-β KL Target (0.05); β 0.005–0.192
entropy (nats)
(d) Length and stop rate
Mean Completion Tokens
340
320
300
280
260
Rollout Stop Rate
1.0
0.8
0.6
0.4
0.2
0.0
GRPO Optimization Step
length (tokens)
stop rate